

REINFORCEMENT LEARNING

LAB ASSIGNMENT - 2

SUBJECT CODE: 24DS742

Submitted by

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IN

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Demo

Train Agent

Number of Episodes

500

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 500)

You won! Reset game?

X

O

O

X

O

X

X

O

Agent's Value Function

0

0

50.0%

0

0

50.0%

50.0%

0

0

RESET GAME

Demo

Train Agent

Number of Episodes

10000

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 10000)

Draw game. Reset game?

X

O

X

O

O

X

O

X

O

Agent's Value Function

0

0

0

0

0

0

0

0

0.0%

RESET GAME

RL Component	Student Observation
Agent	Plays and try to get better at tic-tac-toe
Environment	Tic-tac-toe game with an opponent
State Representation	3 x 3 grid with unmarked or X/O marked cells
Action Space	Mark X/O in an empty cell of the grid
Reward Function	Winning by marking a full column, row or diagonal, Negative if loosing.
Learning Approach Used	Value Function Learning, Temporal Difference

10 games played per episode block.

Training Episodes	Win%	Loss%	Draw%	Behaviour Observed
100	0	80	20	Plays mostly random
500	10	50	40	One strategic move of triangle formation
1000	0	30	70	More draws
5000	20	20	60	Strategic move of exploiting when opponent keeps in a corner.
10000	10	10	80	Plays Optimally

1. Who is the learning agent in this environment?

The agent is the one playing the tic-tac-toe game with another agent or the user. (X/O). It learns how to play the game better by playing many games.

2. What is considered as the environment?

The environment is where the actions of the agent take place which in this case is the tic-tac-toe game and the user/other agent. The game has a 3 x 3 board, X and O markers and rules to be followed.

3. How is the game state represented?

The game state is represented in a 3 x 3 grid of tic-tac-toe board with X and O marked and unmarked empty cells.

4. What are the possible actions available to the agent?

The agent can choose to keep an X or an O (depending on initial selection) in any of the empty cells in the tic-tac-toe board.

5. When does the agent receive a positive reward?

The agent receives a positive reward when its actions leads to a win for the agent. Win in tic-tac-toe game is getting three same markers in a row, column or diagonal.

6. Which Reinforcement Learning approach is used for learning?

The Reinforcement Learning used here is Value Function Learning and Temporal Difference. The agent plays many games and learns the values of the board states. Depending on the epsilon greedy parameter it either chooses to explore or exploit. It performs the action it believes will lead it to having a board state with most value.

7. Is the agent trained using labelled data? Justify your answer.

No the agent is not being trained by explicitly labelled data. Instead it learns using RL. It plays multiple games and optimizes its approach by taking note of which actions lead it to a win. There is no dataset involved.

8. Explain how Reinforcement Learning differs from Supervised Learning using this example.

In Reinforcement Learning the agent learns to make the best move through repeatedly playing the game and coming to conclusions based on how it played each game. It then makes adjustments to its functions to play the next game better. There is no labelled dataset it follows and therefore does not require supervision. Supervised learning on the other hand requires the system to be fed with prior knowledge about the results of the actions it is making. It does not go through with the training in live way like RL.

9. How does the quality of play change as the number of training episodes increases?

The quality of play becomes better as training episodes increase as the agent is able to decide which is the better move to play next.

10. At approximately how many training episodes does the agent begin making intelligent decisions?

The agent seems to make really good and strategic moves by 500.

11. At what stage does the learning appear to converge?

It appears to converge around 10000 episodes. At this range the agent plays near optimally.

12. Why does the win percentage improve with additional training?

The win percentage increases because as the training increases the agent gets to play and see more different board states. Hence it is able to make better moves for a particular given state when playing with the user.

13. Why do most games end in a draw after sufficient training?

After sufficient training the agent plays almost near optimally. In a game of tic-tac-toe if both players make the most optimal moves then the games will always end in draw.

14. What happens if the number of training episodes is too small?

If the number of training episodes is too small the agent has seen very less board states and does not have enough experience. Hence it lets the opponent easily win.

Analysis.

In the beginning the agent plays randomly. Once a game finishes it evaluates the board states it passed through and gives each a score in a way. It uses that to play the next game. If the board state repeats it is now able to make a bit more informed decision. The agent alternates to exploring/making a new different action if it loses more games. The agent gradually plays and trains on a huge number of board states by the end of which it is able to make the best move it learned for almost all given board states. Which is how it better at the game.